

Herbs and Other Botanicals in Cancer Patient Care[†]

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Opinion statement

Non-prescription herbal remedies are commonly used by cancer patients in efforts to control their disease or to manage symptoms associated with cancer and cancer treatments. We address the issues surrounding the use of herbs, herbal compounds, and other botanical agents in the oncology context. Botanicals are biologically active agents that can be useful under appropriate circumstances, but they may be counterproductive when used by patients on chemotherapy or on other prescription medications. Herbs and other botanical agents, despite common public belief, are not benign. They should be understood as unrefined pharmaceuticals, with the capacity to produce physiologic change for better or worse. Indeed, many prescription drugs, chemotherapeutic agents among them, were derived from plants and other natural agents, and the search for additional constituents of plants, animals, and minerals for use as pharmaceutical agents remains an active effort on many fronts. Cautions, appropriate application, and potential utility of botanical agents are discussed below, and sources of reliable information are provided.

Introduction

Botanicals have served as medicines across cultures and throughout time. Knowledge about plants that were found most effective against particular ailments was passed down to succeeding generations. These caches of ancient wisdom encompassed diagnostic techniques, instructions for preparation of remedies, and instructions about which herbs should be prepared with specific other natural products to achieve optimal results. The knowledge and experience of

subsequent generations was added to continually improve these approaches over the centuries, producing increasingly refined medical compendia. These collections are known as Traditional Medical Systems. Ayurveda from India, Traditional Chinese Medicine, and numerous other geographically based systems still thrive in many areas of the world and have given rise to many modern pharmaceutical products.

Advances in science transformed the study and practice of medicine, replacing raw herbs and herbal compounds with powerful synthetic drugs. Magnetic resonance imaging (MRI), blood tests for genetic

[†] Information similar to that included here was presented in the authors' previous publications.

defects, and other sophisticated technology supplanted early diagnostic traditions. While today's high-tech diagnostic and therapeutic advances enable restoration of health for many previously incurable illnesses, including many cancers, they are not uniformly successful. Moreover, chemotherapy and other successful cancer treatments produce adverse effects that are often difficult to control.

These dual cancer-related deficits—less than universal cure rates and successful therapies that nonetheless engender serious and often enduring side effects—lead many cancer patients to look for potential cures outside mainstream medicine, and to seek relief of the distressing symptoms that impair their

quality of life. While modern diagnostic technology has replaced early primitive approaches in much of the world, there is now a resurgence of interest in botanical medicines—those more dilute, less powerful, but less toxic and often effective remnants of humankind's original approach to illness and disease.

This phenomenon also applies to other components of traditional healing systems, such as acupuncture, meditation and physical activity, which comprise the other three papers in this Section. Each utilizes a unique element of the early practice systems. These aspects of Traditional Medical Systems have resurfaced and now hold a place in today's armamentarium against illness and disease.

Prevalence of current botanical use

- Botanicals remain in use by the majority of people worldwide. In developing countries, they are the primary and sometimes the sole means of treating disease, and in the West, botanicals are used most commonly as dietary supplements. Sales of herbal supplements have mushroomed during the last three decades in developed countries. In the United States, according to the 2002 National Health Interview Survey conducted by the Centers for Disease Control and Prevention, about 38 million, or 12% of Americans use botanicals [1].
- Among cancer patients in the U.S., up to 60% use herbal supplements during or after chemotherapy [2–4]. Symptom control, quality of life issues, and cancer recurrence are major driving forces [2, 5]. Research dollars and scientific interest have expanded at least partly in response to active levels of patient use and general public interest. Government agencies and foundations increasingly support studies to evaluate the safety and utility of herbal remedies. Research is essential, as the longevity of a botanical agent or its having served a specific medicinal purpose for centuries does not guarantee safety or efficacy. Indeed, some herbal remedies produce serious adverse effects and should be avoided, as shown in Table 1.

Table 1. Herbal products with potentially serious adverse effects (Source: U.S. Food and Drug Administration)

Product (responsible constituents if known)	Adverse effects
Aristolochia, Bragantia, or Asarum species (aristolochic acid)	Renal toxicity that can lead to renal failure
Common comfrey, prickley comfrey, and Russian comfrey (pyrrolizidine alkaloids)	Hepatic toxicity, veno-occlusive disease
Ephedra, Ma huang (ephedrine alkaloids)	Sympathomimetic activity, hypertension, tachycardia, increased risks for stroke, heart attack, and heart failure
Products containing androstenedione, or "andro"	Androgenic and estrogenic effects
Kava	Hepatic toxicity
St. John's Wort	Potent cytochrome P450 3A4 inducer, altered metabolism of many drugs

Botanical formulations and limitations of current data

- Botanical preparations typically are complex formulations containing multiple ingredients, such as raw herbs, whole or cut plants, isolated plant parts, processed juices, resins, or oils. These may be mixed with yet other materials. The finished product can take a variety of forms, including powder, liquid, pills, or topical creams or gels. Unlike single-molecule prescription medications, botanical agents represent a daunting set of investigational hurdles.
- When and how to apply botanical medicines is based almost exclusively on traditional experience over the centuries, and potential benefits remain largely unstudied in modern terms. Although traditional use is an important basis for investigation, reliance on longterm traditional use is not appropriate in the clinical setting. The biologically active compounds contained in botanicals can cause adverse reactions, some herbs interfere with coagulation or produce other untoward events, and many interact with prescription medications [6, 7]. The popular herb St. John's wort, for example, while an effective treatment for moderate to severe depression [8], induces cytochrome P450 3A4, an enzyme responsible for the metabolism of chemotherapy and many other drugs [9]. Astragalus, an herb commonly used as an immunostimulant in Chinese medicine, can reverse cyclophosphamide-induced immune suppression [10, 11]. Similarly, soy isoflavones may antagonize tamoxifen-associated breast cancer prevention [12]. Because most herbs remain untested, it should be assumed that many are likely to interact with prescription medications.
- As suggested by the St. John's wort and other examples above, the use of botanicals in the context of cancer treatment remains a distinct and important concern. Few studies have addressed the safety and efficacy of botanicals alone or with concurrent chemotherapy, and optimal dosage levels and mechanisms of action remain largely unstudied. Research has begun to address these major gaps only recently, although research endeavors struggle simultaneously to manage the unique set of difficulties inherent in the investigation of botanicals.

Special challenges in botanicals research

- The study of botanicals requires the same high level of research methodology that is applied to the study of any other type of agent, but there are additional challenges unique to botanicals research. An initial hurdle concerns locating the study material itself. Although some study agents are available as commercial products, ready for investigation, others require collection and processing. Plants collected in the wild must be sorted and identified by trained personnel. Their quantity may be limited and future batches may not be entirely consistent with previous batches, or may be unavailable. Raw material captured in the wild presents special difficulties because the properties of a given botanical vary substantially by climate, geographic region, and the time or season of harvest.
- Fortunately, most botanicals used in dietary supplements today are cultivated, rather than harvested in the wild. Optimally, they are farmed using Good Agricultural Practices (GAP) and Good Handling Practices (GHP) as defined by the U.S. Department of Agriculture [13], the United Nations [14], and the European Union [15]. Because the source is known and the future supply of essentially identical material is assured, cultivated specimens are more reliable and preferred.

- Botanicals intended for study should be authenticated by a qualified botanist to confirm consistency with macroscopic and microscopic literature descriptions. It is essential that researchers document and report the specific part of the plant (stem, leaves, flower, roots, etc.) that was used to develop the study agent, as each part contains different bioactive components. It is likely that contradictory conclusions reported by researchers evaluating the same herb are due to use of different parts of the same herb.
- In traditional medical systems, the ingredients, manufacture, and processing of medicinal botanical products were maintained as family or community secrets. Today, although secrecy still remains, research standards require transparency. Specific operating procedures should be noted and pursued, and meticulous documentation in batch records increasingly is adopted as the international standard. Botanical agents are tested for impurities, including residues or solvents that cannot be separated during processing, and contaminants such as pesticides, microbials, heavy metals, or adulterants. Guidelines for the identification, manufacture, and quality control of botanical study agents have been developed in recent years by the NIH [16], the FDA [17], and in a “Consolidated Standards of Reporting Trials” (CONSORT) Statement on Reporting Randomized, Controlled Trials of Herbal Interventions [18].
- Because plants are the source of many successful chemotherapeutic agents, such as paclitaxel, irinotecan, and vinorelbine, serious efforts continue to search for new cancer treatments in herbal medicines. Although in vivo laboratory data often appear promising, pursuant clinical studies often are disappointing, primarily because of bioavailability and pharmacokinetic issues. Achieving effective concentrations of active compounds in target tissues by oral administration is difficult. Another challenge is that time-honored herbal remedies are rarely tested for purity, examined for consistency, or studied carefully.

Current research

- PubMed.gov contains 766 entries for “botanicals or herbs and cancer.” Among them are 28 clinical trials, including 18 randomized studies, most published in Chinese [19]. A total of 88 clinical trials involving botanicals are registered on the NIH Clinical Trials Web site [20]; half are double-blind, placebo-controlled trials.
- Although high-quality science is increasingly applied to the study of botanical agents, not all botanical remedies in common use have been subjected to proper investigation. Available data about some of the most popular remedies are given below. These summaries indicate that properly conducted research shows that at least some historic herbal cancer remedies, pretested to ensure purity and consistency of product and studied with appropriate, rigorous methodology, produce potentially useful, non-toxic cancer treatments. Examples of faulty research, use of contaminated agents, and the persistent popularity of products found to be essentially worthless are also apparent in studies of botanical remedies that count among the most popular in North America today.

Popular botanical agents

- *Essiac*: For decades, this herbal compound has remained a popular herbal therapy for cancer in North America. Developed initially by a Native healer from Southwestern Canada, it was popularized by a Canadian nurse, Rene Caisse (*Essiac* is *Caisse* spelled backward).

- Essiac contains four herbs: burdock, turkey rhubarb, sorrel, and slippery elm. Marketers produce variations on the basic theme by adding additional ingredients. Researchers at the National Cancer Institute and elsewhere found that it has no anticancer effect. Moreover, Essiac was found to stimulate the growth of human breast cancer cells [21].
- *Iscador*, a derivative of mistletoe, is a popular cancer remedy in Europe. It is used in many mainstream European cancer clinics, typically in conjunction with chemotherapy. Despite many studies over the decades, there are no definitive data in support of its value as a cancer treatment.
 - *Mushroom-derived compounds* from Asia show promise in oncology. Several are approved for use as cancer treatments in Japan and there are randomized Phase III trial data that provide evidence of effectiveness. Polysaccharide kureha (PSK), an extract of the mushroom *Coriolus versicolor*, or schizophyllan (SPG), has been subjected to a number of studies. PSK is extracted from the culture medium of *Schizophyllum commune* Fries. Trials typically compared chemotherapy or radiotherapy plus/minus mushroom extract, and found superior survival with PSK as compared to controls in esophageal and gastric cancer patients [22–24].
 - Two randomized trials of PSK extract given after curative resection for colorectal cancer were conducted. In the first ($n = 120$), median survival in the PSK group was approximately 5 years compared with slightly over 4 years in controls [25]. A subsequent trial randomized 462 colorectal cancer patients prior to curative resection to chemotherapy with or without PSK [26]. Disease-free and overall survival were significantly higher in the PSK group. In recent randomized trials, oral PSK with tegafururacil reduced recurrence and increased survival in stage III patients with colorectal cancer [27, 28]. Results are less encouraging in breast cancer [29, 30] and leukemia [31].
 - *PC-SPES* (PC for prostate cancer; SPES is the Latin word for hope), a combination of eight herbs, all but one from traditional Chinese medicine, reduces prostate-specific antigen (PSA) levels in men with advanced prostate cancer regardless of whether disease is androgen-dependent [32–34]. Improvements in quality of life also have been reported [33]. PC-SPES was suspended from the market in early 2002 when undisclosed contamination with anxiolytic and antithrombotic agents was uncovered [35]. A Phase I trial published in 2008 found uncontaminated “PC-Spes2” to be worthy of further investigation as a treatment option for patients with hormone refractory prostate cancer [36].
 - *Sho-saiko-to* (*Xiao-Chai-Hu-Tang*), a traditional Asian medicine for liver disease, comprises seven botanicals. It has marked antiproliferative effects on various cancer lines, particularly hepatoma [37]. In a Phase III randomized trial, 260 patients with cirrhosis were randomized to treatment with Sho-saiko-to or control [38]. At 5 years, Sho-saiko-to led to a one-third reduction in the incidence of hepatocellular carcinoma (23% vs. 34%) and a 40% reduction in deaths (24% vs. 40%). Additional trials are underway.
 - *SV Soup*: A formula of 19 vegetables from traditional Chinese medicine believed to have antitumor activity, called SV Soup, was added to the diets of 12 patients with stages III and IV non-small cell lung cancer. Thirteen clinically similar patients served as controls. All patients received conventional therapies. The median survival of control patients was 4 months (mean 4.8 months), while that of patients who received the vegetable remedy was 15.5 months (mean 15 months). This non-toxic vegetable brew was also reported to improve quality of life [39].

- *Scutellaria barbata*, also known as ban zhi lian and skullcap, is a plant used in traditional Chinese medicine. It demonstrated encouraging clinical activity in a small Phase I trial of patients with advanced breast cancer, but further investigation is needed.
- *Saw palmetto* (*Serenoa repens*) extract is a widely used herbal therapy for BPH. In vitro studies suggest 5 α -reductase inhibition effects [40]. A meta-analysis of 11 randomized trials and two open-label studies of BPH patients on Permixon, a brand of saw palmetto extract, showed that it improved peak flow rate and reduced nocturia [41]. However, another large double blind trial found no significant benefits for saw palmetto on residual voiding volume or serum PSA levels [42]. These conflicting data may be due to the use of different extracts in the studies.
- *Lycopene* is a carotenoid found in abundance in tomato, watermelon, and rose hip. In addition to its antioxidant effects [43], lycopene is a COX-2 inhibitor that increases the expression of connexin 43 [44], which prevents neoplastic transformation. Epidemiologic data support the use of lycopene for prostate cancer [45] and BPH [46], but clinical trial results are mixed [47, 48]. Lycopene is well tolerated, but its clinical benefits await further clarification.
- Soy (*Glycine max*) contains isoflavones, such as genistein and daidzein, which have phytoestrogenic activity. In a case control study of 200 prostate cancer patients and 200 matched controls, intake of isoflavones was significantly associated with decreased risk of prostate cancer ($P < 0.01$). In those with the highest soy intake (≥ 89.9 mg/d), the odds ratio for prevalence of prostate cancer was 0.42 compared to those with the lowest intake of soy (< 30.5 mg/d) [49]. However, in an open-label study of 52 patients with prostate cancer, intake of a genistein-rich extract failed to decrease PSA levels at 6 months [50]. Several studies to determine the benefits of soy in prostate cancer patients are underway.

Conclusions

- Botanical supplement use by cancer patients reflects intense interest and utilization. Fortunately, efforts to subject herbal remedies to rigorous scientific research were initiated in recent decades. Because these agents are biologically active and may interact with chemotherapy and other prescription medications, oncologists should become familiar with supplements in common use and with sources of reliable information. The Memorial Sloan-Kettering Cancer Center "AboutHerbs" Web site, <http://www.mskcc.org/aboutherbs>, provides clinically useful information about herbs and other dietary supplements, with entry portals for oncology professionals and patients. Either portal may be entered at no charge. Other reliable sources are given below in Table 2.

Table 2. Reliable sources of information

Medline Plus: http://www.nlm.nih.gov/medlineplus/druginformation.html
British Medical Journal: http://www.biomedcentral.com/bmccomplementalalternmed/
NIH National Center for Complementary and Alternative Medicine (NCCAM): http://nccam.nih.gov
NCI Office of Complementary and Alternative Medicine (OCCAM): http://dctd.cancer.gov/ProgramPages/OCCAM.htm
NIH Office of Dietary Supplements: http://dietary-supplements.info.nih.gov
Quackwatch/National Council Against Health Fraud: http://www.quackwatch.com/
U.S. Pharmacopeia: http://www.usp.org/dietarySupplements

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